

QLN1 – QLN1 TASK 2: DEVELOPING AND EVALUATING THE NATURAL LANGUAGE PROCESSING MODEL

NATURAL LANGUAGE PROCESSING – D803

PRFA – QLN1

Preparation

Task Overview

Submissions

Evaluation Report

COMPETENCIES

4141.1.2: Applies NLP for Text Analysis and Generation

The learner applies natural language processing techniques for text analysis and generation tasks.

4141.1.3: Optimizes Natural Language Models

The learner optimizes natural language models.

INTRODUCTION

Understanding how to design and optimize a natural language processing (NLP) system is essential for effectively applying machine learning techniques to text data. In this task, you will complete a hands-on project in which you develop an NLP model, document the necessary tools and libraries, implement the system, and evaluate its performance.

As part of this process, you will design a functional NLP model for a specific task, such as sentiment analysis or text classification. You will then document the key libraries and tools required for implementation, using Python as the recommended programming language and PyCharm as the preferred development platform. Once your model is implemented, you will assess its performance using evaluation metrics such as accuracy, precision, recall, F1-score, and a confusion matrix.

To enhance your model's effectiveness, you will apply optimization techniques, including hyperparameter tuning, algorithm selection, and improved data preprocessing. You will analyze the impact of these refinements on your model's accuracy and efficiency, reflecting on best practices for NLP system development.

Additionally, this project can serve as a portfolio piece, showcasing your ability to build, analyze, and optimize real-world NLP models. Throughout the task, you will demonstrate professional communication and adhere to APA (7th edition) citation standards for any referenced materials.

REQUIREMENTS

Your submission must represent your original work and understanding of the course material. Most performance assessment submissions are automatically scanned through the WGU similarity checker.

 Help

Students are strongly encouraged to wait for the similarity report to generate after uploading their work and then review it to ensure Academic Authenticity guidelines are met before submitting the file for evaluation. See [Understanding Similarity Reports](#) for more information.

Grammarly Note:

Professional Communication will be automatically assessed through Grammarly for Education in most performance assessments before a student submits work for evaluation. Students are strongly encouraged to review the Grammarly for Education feedback prior to submitting work for evaluation, as the overall submission will not pass without this aspect passing. See [Use Grammarly for Education Effectively](#) for more information.

Microsoft Files Note:

Write your paper in Microsoft Word (.doc or .docx) unless another Microsoft product, or pdf, is specified in the task directions. Tasks may not be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc. All supporting documentation, such as screenshots and proof of experience, should be collected in a pdf file and submitted separately from the main file. For more information, please see [Computer System and Technology Requirements](#).

- A. Create your subgroup and project in GitLab using the provided "GitLab" and the "GitLab How-To" web links by doing the following:
- Clone the project to the IDE.
 - Commit with a message and push when you complete *each* requirement listed in parts B–G.

GitLab: <https://lrps.wgu.edu/provision/353990238>

GitLab How-To: <https://cm.wgu.edu/t5/Frequently-Asked-Questions/WGU-GitLab-Environment/ta-p/50512>

Note: You may commit and push whenever you want to back up your changes, even if a requirement is not yet complete.

- Submit a copy of the GitLab repository URL in the "Comments to Evaluator" section when you submit this assessment.
- Submit a copy of the repository branch history retrieved from your repository, which must include the commit messages and dates.

Note: Wait until you have completed all the following prompts before you create your copy of the repository branch history.

- B. Design a working model in Gitlab using NLP techniques.

Note: Python and PyCharm are the recommended development language and platform to use for part C.

C. Prepare a write-up describing the libraries and tools necessary for implementing the design.

Note: Requirements Responses to parts D–F should be 2–3 paragraphs each in length and cited by peer-reviewed articles that have been published within the last three years.

D. Describe the implementation process of the NLP system, using parts B and C.

E. Provide an evaluation summary for the performance of the model (e.g., accuracy, precision, recall, F1-score, and confusion matrix).

F. Provide a detailed summary on the optimization of the NLP model (e.g., tune hyperparameters, adjust the architecture, try different algorithms, or use additional techniques like data preparation to improve performance).

G. Acknowledge sources, using APA-formatted in-text citations and references, for content that is quoted, paraphrased, or summarized.

H. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ()

File size limit: 200 MB

File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A:GITLAB

NOT EVIDENT

This submission does not establish a project and subgroup in GitLab.

APPROACHING COMPETENCE

This submission does not accurately establish a project and subgroup in GitLab.

COMPETENT

This submission accurately establishes a project and subgroup in GitLab.

B:WORKING MODEL

NOT EVIDENT

APPROACHING COMPETENCE

COMPETENT

This submission does not provide a working model.

This submission does not accurately provide a working model in Gitlab using NLP techniques.

This submission accurately provides a fully functional model in Gitlab using NLP techniques.

C:WRITE-UP

NOT EVIDENT

This submission does not include a write-up.

APPROACHING COMPETENCE

This submission includes an accurate write-up but lacks 1 or more of the necessary descriptions of libraries or tools.

COMPETENT

This submission includes an accurate write-up describing *all* libraries and tools necessary for implementing the design.

D:NLP SYSTEM

NOT EVIDENT

This submission does not describe the implementation process.

APPROACHING COMPETENCE

This submission does not accurately describe the implementation process for the NLP system, using parts B and C.

COMPETENT

This submission accurately describes the implementation process for the NLP system, using parts B and C.

E:EVALUATION SUMMARY

NOT EVIDENT

This submission does not provide an evaluation summary.

APPROACHING COMPETENCE

This submission does not provide an accurate evaluation summary.

COMPETENT

This submission provides an accurate evaluation summary.

F:SUMMARY OF OPTIMIZATION

NOT EVIDENT

This submission does not provide a summary of optimization.

APPROACHING COMPETENCE

This submission does not provide an accurate summary of optimization.

COMPETENT

This submission provides an accurate summary of optimization.

G:APA SOURCES

NOT EVIDENT

The submission does not include in-text citations and references according to APA style for content that is quoted, paraphrased, or summarized.

APPROACHING COMPETENCE

The submission includes in-text citations and references for content that is quoted, paraphrased, or summarized but does not demonstrate a consistent application of APA style.

COMPETENT

The submission includes in-text citations and references for content that is quoted, paraphrased, or summarized and demonstrates a consistent application of APA style.

H:PROFESSIONAL COMMUNICATION**NOT EVIDENT**

This submission includes pervasive errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation, negatively impacting the professional quality and clarity of the writing. Specific errors have been identified by Grammarly for Education under the Correctness category.

APPROACHING COMPETENCE

This submission includes substantial errors in professional communication related to grammar, sentence fluency, contextual spelling, and/or punctuation. Specific errors have been identified by Grammarly for Education under the Correctness category.

COMPETENT

This submission includes satisfactory use of grammar, sentence fluency, contextual spelling, and punctuation, which promote accurate interpretation and understanding.

WEB LINKS

[Cloud Academy Environment](#)

A sandbox environment for creating the model, before submitting through GitLab

[GitLab](#)[GitLab How-To](#)